MALAWI UNIVERSITY OF BUSINESS AND APPLIED SCIENCES (MUBAS)

SCHOOL OF APPLIED SCIENCES
DEPARTMENT OF COMPUTING AND INFORMATION TECHNOLOGY

Submitted to:

Dr. A. TAYLOR

Presented by:

DANIEL MSAMILA	(BIT/22/SS/020)
ABDUL CARRIM LAIBU	(BIT/22/SS/017)
CALEB PHIRI	(BIT/22/SS/029)
VITUMBIKO KAMANGA	(BIT/22/SS/012)

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Artificial Intelligence (AI)

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NLP (NER, POS Tagging and Speech-to-Text)

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Introduction

This document presents three NLP tests applied to excerpts from the TechCrunch article "In 2026, AI Will Move from Hype to Pragmatism."

Each test demonstrates a key NLP task: Named Entity Recognition (NER), Part of Speech (POS) Tagging, and Speech to Text (STT).

The following Python libraries were used:

- **spaCy (en_core_web_sm model):** for NER and POS tagging
- **NLTK:** for tokenization, POS tagging, and NER chunking
- **SpeechRecognition:** for transcribing spoken audio to text

Test 1: Named Entity Recognition (NER)

Tool: spaCy (en_core_web_sm)

Task: Identify and classify real-world named entities in text (persons, organizations, dates, etc.)

1.1 Input Text

The following excerpt from the article was used as input:

```
If 2025 was the year AI got a vibe check, 2026 will be the year the tech gets practical.
The focus is already shifting away from building ever-larger language models and toward the harder work of making AI usable. Anthropic's Model Context Protocol, a so-called USB-C for AI, lets AI agents talk to external tools like databases, search engines, and APIs.
OpenAI and Microsoft have publicly embraced this standard.
```

1.2 Code Used

```
import spacy

nlp = spacy.load('en_core_web_sm')
```

```
doc = nlp(text1)

for ent in doc.ents:

    print(ent.text, ent.label_, spacy.explain(ent.label_))
```

1.3 Results

Table 1 Below shows the results of the names of the entities recognized.

Entity Text	Label	Meaning
2025	DATE	Absolute or relative dates or periods
2026	DATE	Absolute or relative dates or periods
Anthropic	ORG	Companies, agencies, institutions
OpenAI	ORG	Companies, agencies, institutions
Microsoft	ORG	Companies, agencies, institutions

1.4 Observation

spaCy correctly identified three organizations (Anthropic, OpenAI, Microsoft) and two date references (2025, 2026). The model performed well on technology-domain text, where company names are prominent. It did not misclassify 'AI' or 'USB-C' as entities, showing good contextual understanding.

Test 2: Part-of-Speech (POS) Tagging

Tool: NLTK (averaged_perceptron_tagger)

Task: Assign grammatical roles (noun, verb, adjective, etc.) to each word in a sentence.

2.1 Input Text

If 2025 was the year AI got a vibe check, 2026 will be the year the tech gets practical.

2.2 Code Used

```
from nltk.tokenize import word_tokenize

from nltk import pos_tag

tokens = word_tokenize(sentence)

pos_tags = pos_tag(tokens)

for word, tag in pos_tags:

    print(word, '->', tag)
```

2.3 Results

Table 2 Below shows the results of the Part-of-Speech tagging.

Word	POS Tag	Meaning
2025	CD	Cardinal number
was	VBD	Verb, past tense
year	NN	Noun, singular
AI	NNP	Proper noun, singular
got	VBD	Verb, past tense
vibe	NN	Noun, singular

check	NN	Noun, singular
2026	CD	Cardinal number
will	MD	Modal verb
gets	VBZ	Verb, 3rd person singular present
practical	JJ	Adjective

2.4 Observation

NLTK's POS tagger correctly identified the grammatical structure of the sentence. Notable findings: 'practical' was tagged as JJ (adjective), 'gets' as VBZ (3rd-person present verb), and 'AI' was treated as a proper noun (NNP). Modal verb 'will' was correctly tagged MD, and both years were tagged as cardinal numbers (CD). This demonstrates that NLTK handles technical vocabulary reasonably well, though it lacks the contextual depth of spaCy.

Test 3: Speech-to-Text (STT)

Tool: SpeechRecognition + Google Web Speech API

Task: Convert spoken audio to written text and handle errors gracefully

3.1 Description

This test demonstrates live microphone transcription using the SpeechRecognition library. The system listens for speech input, adjusts for ambient noise, and sends audio to the Google Web Speech API for transcription. Error handling is included for silent input, unclear audio, and API failures.

3.2 Code Used

```
import speech_recognition as sr

recognizer = sr.Recognizer()

with sr.Microphone() as source:

    recognizer.adjust_for_ambient_noise(source, duration=1)

    audio = recognizer.listen(source, timeout=8, phrase_time_limit=10)

text = recognizer.recognize_google(audio)

print('Transcription:', text)
```

3.3 Test Scenario

The uploaded Audio file had the following spoken phrase:

"What if I told you that I love you would you tell me that you love me now would you tell me that you miss me."

3.4 Output (Transcription Result)

The output is that which is a translation of a voice from the uploaded audio file shown in the screenshot below.

Figure 1 above shows the Transcription Results.

3.5 Error Handling Demonstrated

Scenario	Error Type	System Response
No speech detected in time	WaitTimeoutError	Prints warning; system does not crash
Audio too unclear to understand	UnknownValueError	Prints warning; handled gracefully
No internet / API failure	RequestError	Prints API error message; continues
No microphone found	OSError	Prints hardware warning; skips test

3.6 Observation

The SpeechRecognition library works effectively when used together the Google Web Speech API and a functioning microphone.

Key observations:

- accuracy depends on microphone quality and background noise.
- comprehensive error handling is essential for real-world STT systems.
- An active internet connection is required since Google's API processes audio in the cloud.

Summary of All Tests

Test	Task	Tool	Key Finding

Test 1	Named Entity Recognition	spaCy	Detected 3 ORG (Anthropic, OpenAI, Microsoft) and 2 DATE entities
Test 2	POS Tagging	NLTK	Correctly tagged 'practical' as JJ, 'gets' as VBZ, 'AI' as NNP
Test 3	Speech-to-Text	SpeechRecognition	Transcribed spoken audio; all error scenarios handled gracefully

Conclusion

The three tests demonstrate the practical application of NLP tools on a real technology article:

1. spaCy provides rich, detailed NER output with built-in label explanations and excels in identifying organizational and date entities in tech-domain text.
2. NLTK is a lightweight, modular toolkit that accurately assigns grammatical roles to words, making it ideal for structural analysis of sentences.
3. SpeechRecognition enables effective transcription of spoken language, but requires an internet connection and proper error handling for robust real-world deployment.